n2edm

September 2021

1 Introduction

2 Sums

We measure $B1 = B + B_{Offset}$ and $B2 = -B + B_{Offset}$ by changing the current direction in layer 6.B describes here the magnetic field created by the coils and B_{offset} consists of the offset from the magnetic field sensor and the general surrounding. The sum of measurement values should give 2 B_{Offset} and can be used to study the offset of the magnetic field that cannot be controlled by degaussing. The generator offset can be used to compensate for the constant magnetic field of the surrounding, so that we effectively measure the offset originating from the sensor. For the right Generator offset we expect to measure a uniform distribution of B_{Offset}.

2.1 Projection onto Rho-Phi plane

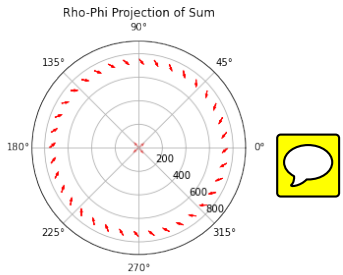

Figure 1: Generator offset = -91 ppm

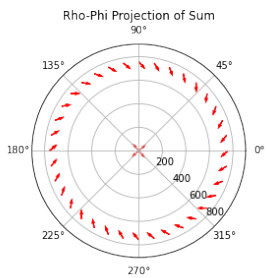

Figure 2: Generator offset = -140 ppm

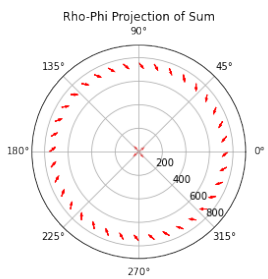

Figure 3: Generator offset = -30 ppm

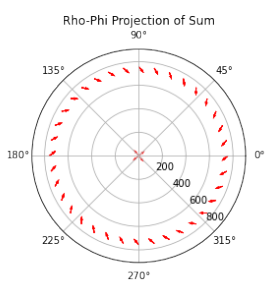

Figure 4: Generator offset = -70 ppm

2.2 Phi component

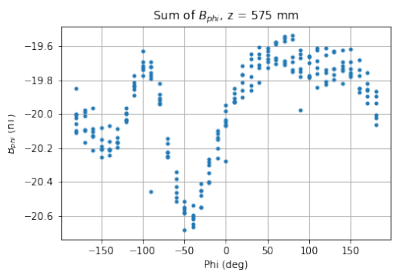

Figure 5: Generator offset = -91 ppm

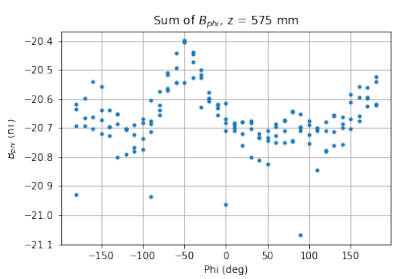

Figure 6: Generator offset = -140 ppm

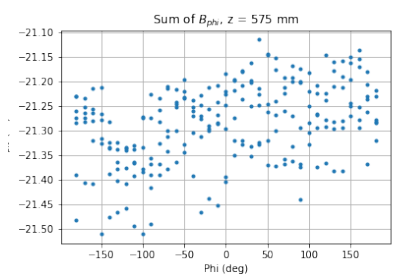

Figure 7: Generator offset = -30 ppm

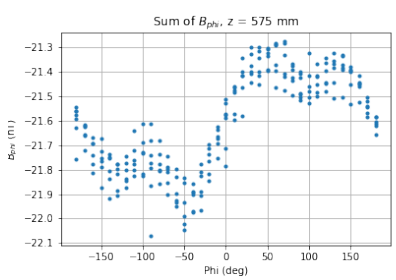

Figure 8: Generator offset = -70 ppm

2.3 Rho component

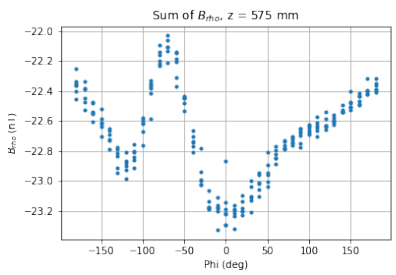

Figure 9: Generator offset = -91 ppm

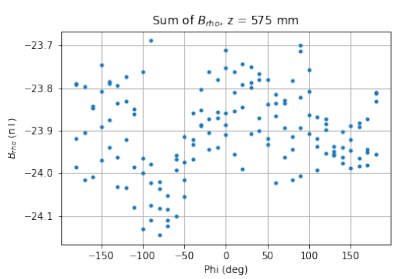

Figure 10: Generator offset = -140 ppm

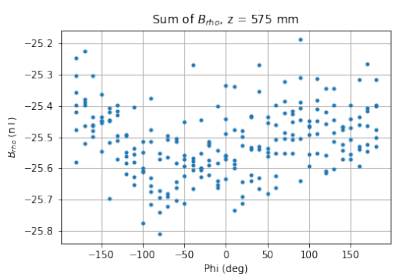

Figure 11: Generator offset = -30 ppm

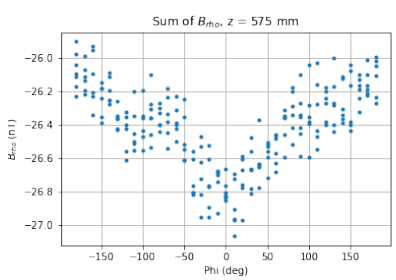

Figure 12: Generator offset = -70 ppm

2.4 z component

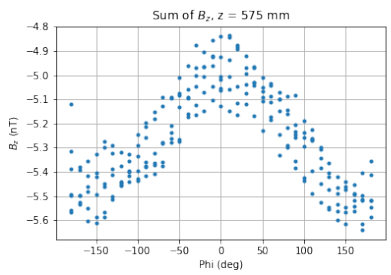

Figure 13: Generator offset = -91 ppm

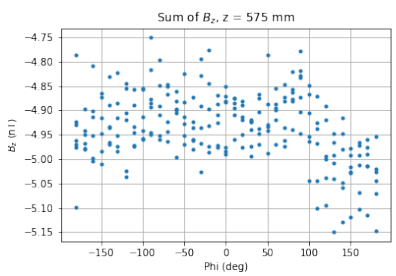

Figure 14: Generator offset = -30 ppm

3 Differences

The Difference of the measurement values should give $2B$.

3.1 Projection onto Rho-Phi plane

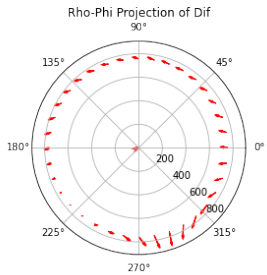

Figure 15: Generator offset = -91 ppm

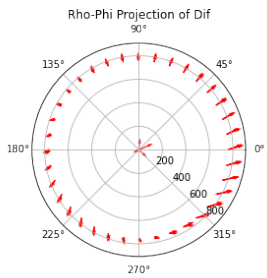

Figure 16: Generator offset = -140 ppm

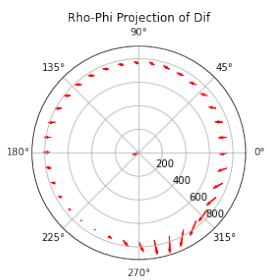

Figure 17: Generator offset = -30 ppm

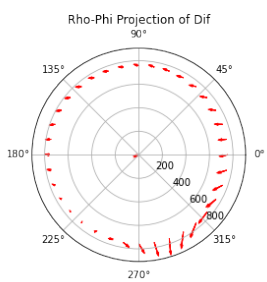

Figure 18: Generator offset = -70 ppm

3.2 Phi component

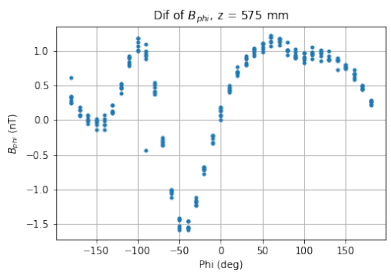

Figure 19: Generator offset = -91 ppm

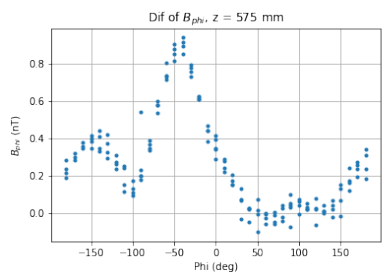

Figure 20: Generator offset = -140 ppm

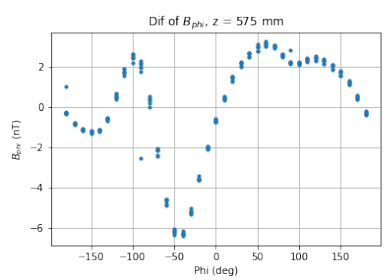

Figure 21: Generator offset = -30 ppm

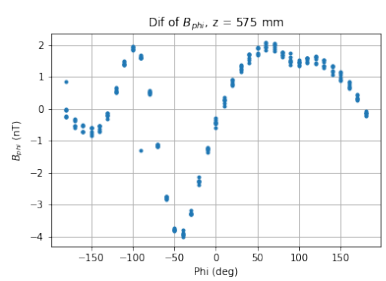

Figure 22: Generator offset = -70 ppm

3.3 Rho component

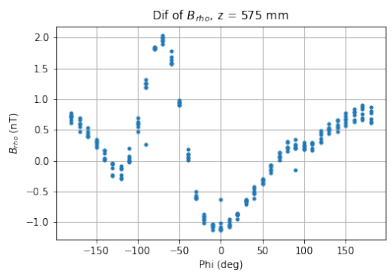

Figure 23: Generator offset = -91 ppm

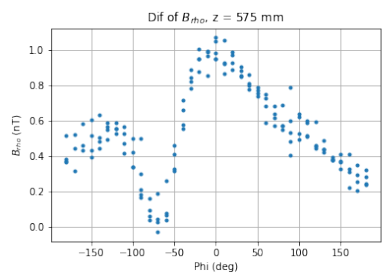

Figure 24: Generator offset = -140 ppm

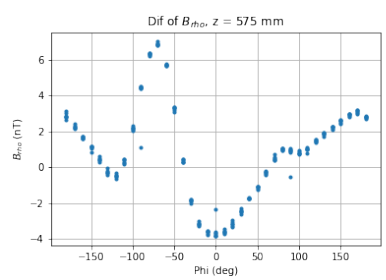

Figure 25: Generator offset = -30 ppm

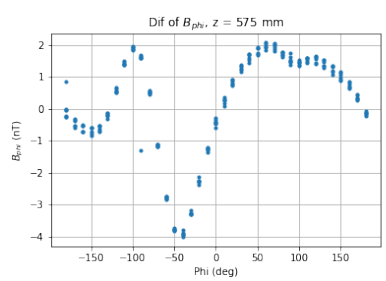

Figure 26: Generator offset = -70 ppm

3.4 z component

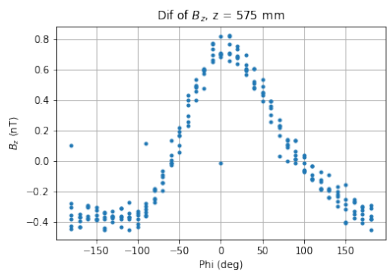

Figure 27: Generator offset = -91 ppm

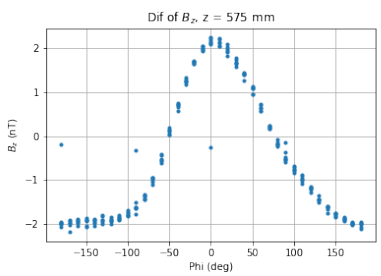

Figure 28: Generator offset = -30 ppm